

Errata

Information Propagation Methods for Reliability, Reachability and Flow Analysis in Directed Acyclic Process Networks

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After submission, two corrections were made to the thesis. The first corrects a statement about minimum cuts in Chapter 6; no number reported in the thesis changes. The second corrects the capacity results of the Net3 case study in Chapter 10, and the statements that depend on them in Chapter 11 and Appendix B. A third section lists minor corrections of description, none of which changes a result. Page numbers refer to the submitted copy, and a copy of the corrected thesis with every change marked (deletions struck through in red, additions underlined in blue) accompanies this sheet. The reliability and schedule results of Chapter 10, and every result in the other chapters, are unchanged. Every corrected value below was computed with version 0.2.3 of the registered package, `InformationPropagationAnalysis.jl`, which contains the Chapter 6 correction.

1. Chapter 6: the structure of the minimum cuts

Section 6.5.1 (pp. 105–106) stated that every node of the free zone $\mathcal{F}$ can be assigned to either side of a minimum cut without changing its capacity, so that there are exactly $2^{|\mathcal{F}|}$ minimum cuts. The converse is not true for every subset. By Picard and Queyranne (1980), $S^* \cup R$ with $R \subseteq \mathcal{F}$ is a minimum cut if and only if R is closed under reachability in the residual graph G_f . There are therefore at most $2^{|\mathcal{F}|}$ minimum cuts, with equality only when no node of $\mathcal{F}$ reaches another in G_f . For example, the power network of Appendix A with every capacity set to 60 has $\mathcal{F} = \{19, 20, 21\}$ linked as a chain in G_f , and 4 of its 8 subsets are minimum cuts.

Location	Correction
Def. 6.3 and 6.4, p. 105	Taking the whole of S^{**} as the source side gives a minimum cut; the claim that any free-zone node may be moved alone is removed. A theorem stating the Picard–Queyranne characterisation is added.
p. 106, degeneracy	“exactly $2^{ \mathcal{F} }$ equivalent minimum cuts” becomes “at most $2^{ \mathcal{F} }$, one for each closed subset”.
p. 106, E_{some}	The condition $v \notin \text{Reach}_{G_f}(u)$ is added: a saturated edge lies in some minimum cut only if its head is not reachable from its tail in G_f .
p. 106, enumerator	The enumerator examines the subsets of $\mathcal{F}$ and keeps those whose cut capacity equals F^* ; <code>is_complete</code> refers to the subsets examined.
Table 6.2, p. 114	Rows <code>edges_some</code> ($O(E_{\text{sat}} (V + E))$, residual reach) and <code>enumerate</code> (closed subsets of $\mathcal{F}$) updated.

The implementation was corrected to match, and the results of Chapter 6 were re-run. Configuration A still has a free zone of two nodes and four minimum cuts, of 13, 8, 8 and 3 crossing edges, and Configuration B one. The IEEE RTS-24 free zone is empty. No reported value changes.

2. Chapter 10: capacity results of the Net3 case study

Section 10.3.2 specifies a super-sink with one edge from each of the 58 demand junctions, with capacity equal to that junction’s demand, 680.1 litres per second in total. The capacity analysis reported in the thesis was run on the edge list without the super-sink and its 58 edges. The reported throughput, 1,837.9 litres per second, therefore exceeded the total demand. The analysis algorithms of the package were not at fault: they computed the maximum flow of the graph they were given. The error lay in the interface layer. The case study’s edge list omitted the super-sink, and the server through which the interface reaches the package (Section 8.8, p. 162) ignored the 58 capacity entries for edges it did not contain, without reporting them. The analysis was re-run on the edge list with the super-sink added and

declared as the only sink, and the results were checked against an independent max-flow computation (Edmonds–Karp, separate implementation), which agrees to the reported precision. The server now rejects a capacity file that names an edge the edge list does not contain.

Quantity	Submitted	Corrected
Baseline throughput	1,837.9 L/s	680.1 L/s, the whole demand
Baseline minimum cut	unique; four edges: two discharges at the river pump’s junction, one at the lake’s, and the source tank outlet	unique (free zone empty); the 58 demand edges
Saturated edges; paths	7; 158	60; 401
Sensitivity, baseline	the four cut edges, 883.3 to 76.0 L/s; edge 95–96 marginal range unbounded	the demand edges, ranked by demand: junction 58, 280.1 L/s; junction 92, 103.3 L/s; every marginal range finite
Degraded throughput	954.6 L/s, a 48% drop	43.3 L/s, 6.4% of demand
Degraded minimum cuts	the same four edges	24 cuts, free zone of six nodes; in every cut, the river pump (95–97) and the demand edges of junctions 7, 8, 77, 79, 80, 81
Flow timing, Table B.5	2.33 s	4.07 s

Interpretation. At baseline the network meets its whole demand, so the network does not bound delivery; the submitted text attributed the bound to the pumps and the source tank (p. 189). In the degraded scenario only the six junctions downstream of the lake pump and the discharging tank in the oriented graph can be served, and the degraded trunk main does not bind, because the river pump is its only feed. This result holds with every pipe in its snapshot direction; a degraded state would need an orientation of its own, and this is stated as a limitation of the operating-snapshot assumption.

Profile overlay. Section 10.7 (pp. 192–195) stated that the minimum cut and the critical path have no node in common. The corrected figure draws the edges in every minimum cut of the degraded scenario against the baseline critical path. They share nodes 95 and 97, the ends of the river pump, and the super-sink. The pump whose loss leaves 6.4 per cent of demand served lies on the restoration critical chain. The submitted overlay also shared nodes 95 and 97; the interface counted the overlap of an edge set by node only, and this is corrected.

Location	Change
§10.1, p. 182	“uploaded once” removed; the capacity analysis reads the edge list with the super-sink added.
§10.3, pp. 183–184	The super-sink is added to the capacity edge list (98 nodes, 177 edges) and named as the only sink.
§10.5, pp. 189–191; Figs. 10.6–10.9	Rewritten with the corrected results; figures retaken from the corrected run.
§10.7, pp. 192–195; Fig. 10.14	Rewritten as above; figure retaken.
§10.8, p. 194	Flow timing 4.1 s.
§10.9, p. 196; §11.2.1, p. 198	Corrected numbers; “from one upload” becomes “from one conversion”.
§11.6, p. 204	The closing remark now states that the pump which bounds throughput in the degraded state lies on the restoration critical chain, in place of “different components”.
Table B.5, p. 220	Flow rows corrected.

3. Minor corrections of description

Location	Correction
App. A, §A.6, p. 209; App. B, §B.4, p. 220 §5.4, p. 65	The PSPLIB interval scenario applies a $\pm 20\%$ relative half-width to every duration, not $\pm 10\%$. Chapter 7 and every reported value (project range [30.4, 45.6]) are for $\pm 20\%$. “ 2^{10} component states” becomes “ 2^9 states of the uncertain components”: the diamond has four uncertain nodes and five uncertain edges (the source has prior 1).
§10.4, p. 186	Node 95 is the junction at the <i>inlet</i> of the river pump, where the river reservoir enters the network, not the junction downstream of it.
§10.3, p. 184	The 1.5 m/s pipe velocity is an assumed design value; the reference to a convention in the section is removed.
§8.3, p. 154	The facade re-exports eleven names (five modules, two value types, four entry points), not twelve.
§9.5, p. 176	The workspace holds nine projects (the shell, five feature libraries, three shared libraries), not eight.
App. A, §A.2, p. 208	Three published-reliability scenario folders, not five.
Table B.1, p. 217	The power-network residual is described as a gap between the topology used and the published values, not as a modelling difference; its cause is unresolved. The reproduction was additionally checked by exhaustive enumeration of all 2^{27} link states, which agrees with the reported values.
Table B.5, p. 220	“certified requirement range” becomes “belief range at node 76”.